

Challenge (Habanero)

- Q1.** A hiker is traveling through a national park. The height h of a mountain in miles above sea level is given by $h = -0.05d^2 + 2d + 1$, where d is the distance from the park entrance. Another mountain's height is given by $h = 0.03d^2 - d + 3$. If the two mountains have the same height, what is their height in miles?
- (A) $h = 4.5$ miles
(B) $h = 5.2$ miles
(C) $h = 4.1$ miles
(D) $h = 3.8$ miles
(E) $h = 5.5$ miles
- Q2.** Solve the system of equations: $x^2 + 2y^2 = 4$ and $x^2 - 2y^2 = -1$. What is the value of x^2 ?
- (A) $x^2 = 1.5$
(B) $x^2 = 2$
(C) $x^2 = 2.5$
(D) $x^2 = 3$
(E) $x^2 = 1$
- Q3.** A park ranger tracks the population P of rabbits and the population S of squirrels. The equations modeling their populations are $P = 0.2t^2 + 3t + 10$ and $S = 0.1t^2 - t + 5$, where t is time in months. After how many months will the rabbit and squirrel populations be equal?
- (A) $t = 5$ months
(B) $t = 3$ months
(C) $t = 2$ months
(D) $t = 7$ months
(E) $t = 4$ months
- Q4.** The number of gallons G of fuel consumed by a car is related to the speed s of the car by $G = 0.01s^2 + 2s + 10$. The distance d traveled is related to the speed by $d = 0.1s^2 - s + 20$. Find the speed s at which $G = 25$ gallons and $d = 15$ miles.
- (A) $s = 30$ mph
(B) $s = 40$ mph
(C) $s = 20$ mph
(D) $s = 25$ mph
(E) $s = 50$ mph
- Q5.** If $x^2 - 4y^2 = 1$ and $x^2 + 2y^2 = 9$, what is the value of x ?
- (A) $x = pm2$
(B) $x = pm1$
(C) $x = pm3$
(D) $x = pm4$
(E) $x = pm5$
- Q6.** A conservation team tracks the population B of bears and the population W of wolves. The equations for their populations are $B = 0.3t^2 - t + 20$ and $W = 0.2t^2 + 2t + 10$. After how many months will the bear and wolf populations be equal?
- (A) $t = 2$ months
(B) $t = 4$ months
(C) $t = 6$ months
(D) $t = 8$ months
(E) $t = 10$ months
- Q7.** The depth D of water in a lake is given by $D = 0.05x^2 - 2x + 10$, where x is the distance from the shore. The volume V of water in the lake is $V = 0.1x^2 + x + 5$. If $V = 20$ cubic meters, find x .
- (A) $x = 5$ meters
(B) $x = 10$ meters
(C) $x = 3$ meters
(D) $x = 8$ meters
(E) $x = 2$ meters
- Q8.** The height h of a waterfall is $h = -0.1d^2 + 3d + 2$, where d is the distance from the viewer. The width w of the waterfall is given by $w = 0.2d^2 - d + 1$. If $w = 2$ meters, what is h ?
- (A) $h = 6$ meters
(B) $h = 5$ meters
(C) $h = 4$ meters
(D) $h = 7$ meters
(E) $h = 3$ meters

Open Ended Questions (FRQ)

- Q9.** Solve the system of equations: $y = x^2 - 3x + 2$ and $y = 2x^2 + x - 1$. Find the points of intersection and explain their significance in the context of a hiking trail.
- Q10.** Graph the equations $y = x^2 + 2x - 1$ and $y = -x^2 + 4x - 3$ on the same coordinate plane. Interpret the points of intersection in terms of the population of rabbits and squirrels in a forest ecosystem.

Chef's Tips

- Always review key concepts before each test.
- Emphasize the importance of showing work and explaining answers.
- Provide additional support for students struggling with quadratic equations.